

Evaluation of antioxidative, analgesic and anti-inflammatory activities of methanolic extract of *Myrica nagi* leaves - an animal model approach

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Abstract *Myrica nagi* (family Myricaceae) is commonly known as *Kathphal* (Hindi) and *Bayberry* (English) and it has a long history of usage in traditional medicine. It is popular actinorhizal plant for its symbiotic relationship with *Frankia*. This study was taken in force to estimate the analgesic, anti-inflammatory and anti-oxidative activities of methanolic extract of *Myrica nagi* (MMN) in an animal model. Anti-oxidative property of MMN was assessed by free radical scavenging assay (DPPH method). The acute toxicity test of methanolic extract of MMN revealed that the median lethal dose (LD50) was found to be 2080 mg/kg body weight in mice. The anti-inflammatory property was evaluated by carrageenan-induced acute inflammation in rats by measuring rat paw volume at different time intervals and toxicological analysis using mice. The analgesic effect was measured in Wistar rats using the acetic acid-induced writhing test and

MMN at 200 mg/kg BW showed 54.56 % inhibition of writhing. MMN showed higher anti-oxidant activity in DPPH assays as compared to standard. High dose of MMN showed a significant reduction (21.71 %) in inflammation after 4 h of treatment, which was comparable to diclofenac (10 mg/kg BW; 32.75 %)-treated group. Significant reduction ($p < 0.05$) in the levels of inflammatory cytokine (IL-1 β and TNF- α) markers were also observed in serum of MMN-treated animals as compared to control. Taken together, the phenolic compounds of MMN may serve as potential herbal drug for amelioration of acute inflammation due to their modulatory action on free radicals.

Keywords Acute inflammation · Analgesic · Anti-oxidant · DPPH · *Myrica nagi*

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1 Introduction

Myrica nagi Thunb. (family Myricaceae) popularly known as *Bayberry* is variously referred as *Kathphal* (Hindi), *Sohphi* (Khasi) and *Nagatenga* (Assamese). *M. nagi* is an actinorhizal plant. Actinorhizal plants are well known for their ability to form root nodules in symbiotic association with the soil actinomycete *Frankia* and are capable of high rates of nitrogen fixation, comparable to those of legumes. It is mostly evergreen, dioecious and is distributed in the subtropical Himalaya from Ravi, eastwards to Assam and in Khasi Hills, Jaintia Hills, Naga Hills and Lushai Hills at an elevation of about 900–1200 m above mean sea level (Panthari et al. 2012). In Meghalaya, three different morphotypes of *Myrica* have been reported viz. *M. esculenta*, *M. nagi* and the third intermediate between the two (Yanthan and Mishra 2013).

Inflammation is a host defence mechanism which is in response to noxious stimuli. It is characterised by redness of

skin, pain, warmth swelling and loss of function. Inflammation plays an important role in various diseases such as rheumatoid arthritis, ulcer, atherosclerosis, and asthma. Many drugs that are available in the market for inflammatory diseases exhibit several side effects to humans. Therefore, there is a need for alternative treatments with safer compounds. Every part of *M. nagi* is precious in Ayurveda as medicine for combating several ailments since ancient times (Silva et al. 2015). Anti inflammatory activity of ethyl acetate and aqueous extracts of bark of *M. nagi* at 100 and 200 mg/Kg BW using carrageenan and histamine induced rat paw edema exhibited about 28 and 18 % in wistar rats. While information is available on the medicinal property of *M. nagi* bark (Khan et al. 2008; Patel et al. 2011; Panthari et al. 2012) and fruits (Mannello 2006), such information is severely lacking for the leaves. This study therefore investigates the leaves of *M. nagi* for their anti-inflammatory, analgesic and anti-oxidative activities. To our knowledge, the current investigations will be the first report on analgesic, anti-oxidative and anti-inflammatory properties of *M. nagi* leaves so far.

2 Materials and methods

2.1 Plant material collection and extraction

M. nagi leaves were collected from the fields of North Eastern Hill University (NEHU) Campus, Shillong, Meghalaya, India during February, 2014. The plant material was authenticated by a plant taxonomist and a voucher specimen (Voucher No. 9659) was deposited at Department of Botany, University of North Bengal, West Bengal, India. The material was shade dried and ground to fine powder. The pulverized leaves (10 g each) were extracted by Soxhlation using methanol (plant: solvent 1:15 w/v) (Middha et al. 2011) for 6 h. The methanolic extract of *M. nagi* leaves (MMN) was dried with the help of a rotary vacuum evaporator (Middha et al. 2011) and stored at 4 °C until required.

2.2 In vitro antioxidant property (DPPH method)

The antioxidant activity of MMN and the standard were assessed on the basis of the scavenging effect of the stable 2, 2-diphenyl-1-picryl-hydrazyl (DPPH) free radical as per the modified protocol by Goyal and his co-workers (Goyal et al. 2010). The DPPH solution (0.006 % w/v) was prepared in 95 % methanol. Freshly prepared DPPH solution was added to all test tubes. The extract was pipetted out in different aliquots (100–1000 µg) and the final volume was made up to 2 mL. The discoloration of the reaction mixture was measured at wavelength 517 nm after 30 min of incubation in the dark. The experiment was performed at least in triplicates. Ascorbic acid was used as a reference standard and dissolved in double

distilled water (DDW) to make the stock solution with the same concentration (1 mg/mL). The solvent methanol (95 %) was used as blank. Percentage scavenging of the DPPH free radical was measured using the equation:

$$\text{Percentage scavenging} = A_0 - A_1 / A_0 \times 100$$

where A_0 is the absorbance of the control and A_1 is the absorbance of the sample. The actual decrease in absorption induced by the test extract was compared with the positive control. The IC₅₀ value was calculated using the dose inhibition curve. IC₅₀ is the desired drug concentration requires producing 50 % inhibition.

2.3 In vivo assays

2.3.1 Animals

Male Wistar albino rats (140–160 g) were housed under standard laboratory conditions of light and dark cycles of 7:00 am to 7:00 pm, temperature of 25 °C ± 2 °C and 68 % ± 1 % relative humidity. The animals were provided standard rat pellet (Lipton India Ltd., Bangalore, India) and tap water *ad libitum*. All animal procedures were carried out in strict compliance with the Institutional Animal Ethics Committee regulations. The study protocol was approved by Maharani Lakshmi Ammanni College Ethical Committee, clearance from ethical committee (1368/ac/10/CPCSEA), Bangalore, India.

2.3.2 Acute toxicity test

Swiss albino mice (25–30 g) of both sexes were divided into six groups of 10 each. Animals were fasted overnight and were free to access water preceding the experiment. MMN was administered to each group at different dose levels of 0.5, 1.0, 1.5, 2.0, 2.5 and 3.0 g/kg Body weight (BW)/mL. The mice were critically observed for 24 h, mortality was recorded and LD₅₀ (median lethal dose) was determined as per previous study (Middha et al. 2011).

2.3.3 Analgesic activity

Acetic acid-induced writhing test: The writhing test for analgesic activity was performed according to the method described by Mohanty et al. (2015). The Wister albino rats were divided in to five groups with six randomly selected animals in each group. All animals were fasted 24 h prior to start of the experiment and had free access to water. Group I was considered normal treated with water alone (NL), group II was treated as the control administered with vehicle only (acetic acid 1 % v/v), group III and IV received low dosage of *M. nagi* methanolic leaf extract (LMMN) and high dosage of *M. nagi* methanolic leaf extract (HMMN) at 100 and 200 mg/Kg body

weight (BW) respectively, and group V was given with the standard drug diclofenac sodium (10 mg/Kg; Sigma Chemical Co., St. Louis, MO, USA). After 1 h of treatment, writhing was induced by intra-peritoneal injection of 1 % acetic acid (1 mL/Kg BW). The writhing response was observed by counting the total number of writhes after 5 min of injection of acetic acid during 20 min. Data were recorded as mean number of writhes from each group and the percentage of inhibition was calculated according to the following formula.

$$\% \text{ inhibition} = \{(W_c - W_t) \times 100\} / W_c$$

where W_c is average number of writhing in control group and W_t is average number of writhing in treated group.

2.3.4 Anti-inflammatory activity

The carrageenan-induced rat paw edema test (Winter et al. 1962) has been used as an experimental model for testing the anti-inflammatory activity as per previous report (Mohanty et al. 2015). One hour after oral administration of LMMN and HMMN or diclofenac, carrageenan-saline solution (0.5 % w/v) and saline were injected in a volume of 0.1 mL into the plantar surface of the right and left hind paw, respectively. The left paw served as the control (non-inflamed) paw. The experimental animals were observed for 0 h, 1 h, 2 h and 4 h; the paw volume was measured using a plethysmometer. The anti-inflammatory effect was calculated as per the equation below:

$$\text{Anti-inflammatory effect}(\%) = k - e / k \times 100$$

where k is the difference in the paw weight in the control group and e is the difference in the paw weight in the treatment group. After treatments, rats were euthanized using combination of 0.15 mL/100 g xylazine (30 mg/kg BW) and 0.3 mL/100 g i.p ketamine (300 mg/kg BW) as proposed by Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA). Blood samples were collected by retro-orbital puncture from all rats and serum was separated for further biochemical estimations.

2.3.5 Measurement of IL-1 β and TNF- α levels

The serum concentrations of interleukin-1 beta (IL-1 β) and tumor necrosis factor-alpha (TNF- α) (Endogen, Woburn, MA, USA) were measured using a commercial enzyme-linked immunosorbent assay (ELISA) kit method (Biosource, San Diego, CA, USA) (Comalada et al. 2005).

2.4 Statistical analysis

All the experiments were repeated six times and expressed as mean $\pm$ standard error of means (SEM)

and statistically analyzed by two-way analysis of variance followed by Tukey's multiple range using GraphPad prism 5. $p < 0.005$ and $p < 0.01$ were considered to be statistically significant.

3 Results and discussion

3.1 DPPH scavenging activity of *M. nagi*

The radical scavenging activity, using a DPPH generated radical, was tested at different concentrations ranging from 10 to 100 mg/ml for MMN (Fig. 1). The values for maximum % inhibition at 10 to 100 mg/ml ranged from 17.15 to 67.34 % in MMN, whereas that of standard (ascorbic acid) was found to be 11.75 and 34.23 % for 10 and 100 mg/ml concentration, respectively. The high DPPH scavenging activities of the methanolic extract have already been reported by other researchers in other plants (Goyal et al. 2011a; Goyal et al. 2011b). IC₅₀ of MMN was found to be 3.34 mg/ml.

DPPH is a stable free radical and its activity is based on its ability to decolorize from deep purple colour at a wavelength 517 nm to light yellow in the presence of antioxidants forming a stable DPPH molecule (Goyal et al. 2013). MMN showed an effective radical scavenging ability by this assay.

3.2 Acute toxicity

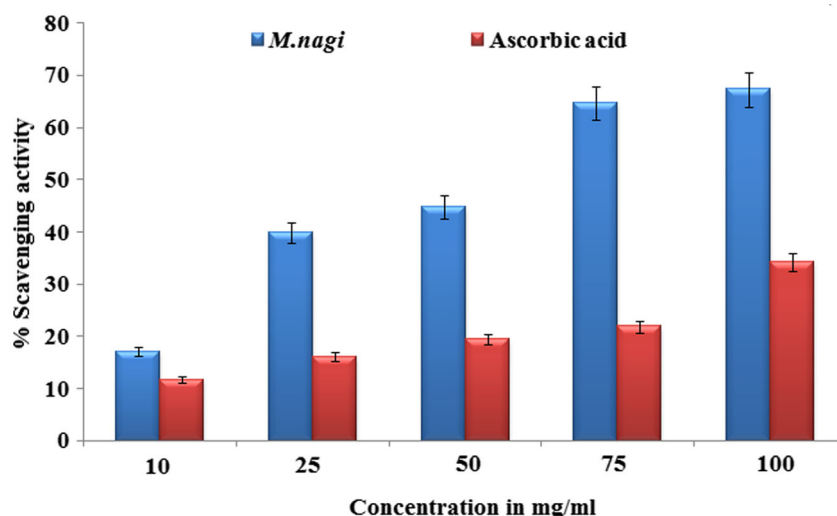
In vivo median lethal dose or LD₅₀ of MMN was 2080 mg/kg BW by oral route. Other parameters such as body weight and survival rate showed no side effects to MMN (data not shown). LD₅₀ is a parameter to calculate toxicity of a sample that kills 50 % of the population and expressed in mg/ml per kg BW.

In vivo toxicological estimation was done to evaluate the extent to which a drug can cause deleterious effects to the cell; these tests are essential and widely accepted to determine optimum drug concentration to be administered. LD₅₀ of MMN was measured to check lethal toxicity. Treatment with MMN at a dose of 100 mg/kg and 200 mg/kg to their respective group of mice for 1 month did not show any toxic side effect.

3.3 Analgesic activity

The results depicted in Table 1 showed that both the doses (100 mg/kg and 200 mg/kg) of MMN were effective in pain relief in a dose-dependent manner. After oral administration of the extract at both the doses the maximum inhibition was observed (77.09 %) in the higher dose i.e. 200 mg/kg of MMN. However, the

Fig. 1 DPPH scavenging activity of methanolic extract of *Myrica nagi* leaves



inhibitory effect of the standard drug diclofenac was found to be greater than both the doses.

Acetic acid-induced writhing involving supraspinal nociceptors signifies non-steroidal anti-inflammatory drug (NSAID) type of analgesia (Vedhanayaki et al. 2003). It is a well-documented protocol for evaluating analgesic activity (Mohanthy et al. 2015). The sensation of pain is elicited by invigorating some pain mediators like arachidonic acid from tissue phospholipid via cyclooxygenase (COX) (Duarte et al. 1988; Ahmed et al. 2006). This event might be in conjunction with the elevated level of prostaglandin E2 and prostaglandin F2 α along with lipoxygenase products in the peritoneal fluids which in turn augments inflammatory pain by increasing capillary permeability (Zakaria et al. 2008). The possible mechanism by which an agent will suppress the writhings rendering analgesic effect is probably by inhibition of prostaglandin synthesis, a peripheral mechanism of pain inhibition (Zulfiker et al. 2010). In our study, prominent inhibition of writhing reflex was observed in both the doses of *M. nagi* methanolic leaf extracts. The significant reduction in pain might be at-

tributed to the analgesic ingredients present in *M. nagi* leaves acting within the prostaglandin pathways or indirectly by releasing pain mediators.

3.4 Anti-inflammatory activity

3.4.1 Carrageenan-induced paw edema

A number of experimental methods are employed to access the anti-inflammatory activity. We opted for carrageenan-induced paw edema model since it is considered to be the most suitable and widely used test system for anti-inflammatory activity of natural products (Viana et al. 2011). Being a biphasic process of 2 h each, the carrageenan-induced edema is characterized by initial release of histamine, serotonin in the first 2 h followed by the release of prostaglandin, bradykinins, leukotrienes and kinin in the later stage (Usha et al. 2014). The mean increase in paw volume and percentage of edema inhibition from 0 h up to 4 h with different concentrations of the extract, standard and control is represented in Table 2. The result re-

Table 1 Analgesic activity by methanolic extract of *M. nagi* leaves in experimental rats

Experimental Animals	Dosages (mg/kg)	No of Writhing	Percent Inhibition
NL	No induction	2.60 $\pm$ 0.09	—
AA(1%v/v)	No treatment	41.13 $\pm$ 4.56 ^a	—
AA + LMMN	100 mg/kg	42.17 $\pm$ 2.56 ^a	-2.46 $\uparrow$
AA + HMMN	200 mg/kg	23.16 $\pm$ 2.20 ^b	+77.59 $\downarrow$
AA + Diclofenac	10 mg/kg	4.80 $\pm$ 0.18 ^c	+177.09 $\downarrow$

NC, Normal Control; AA, Control Acetic Acid (1%v/v); AA + LMMN, Low MN (100 mg/Kg body weight); AA + HMMN, High MN (200 mg/Kg body weight). Data represented as mean $\pm$ SEM for number of animals = 6. * p < 0.05 (one way ANOVA and Tukey's test). Different subscripts denotes significant group at p < 0.05 vs. inflammatory group (IC); NS indicates not significant

Table 2 Paw volume variation due to anti-inflammatory effects of methanolic extracts of *M. nagi* leaves

Experimental Animals	Dosages mg/kg	% increase 1 st hour	% increase 2 nd hour	% increase 4 th hour
NL	—	2.60 ± 0.09	2.90 ± 0.01	2.20 ± 0.01
IC	No treatment	41.13 ± 4.56	39.21 ± 9.01 ^a	62.48 ± 10.14 ^a
IC + LMMN	100	42.76 ± 1.93 ^{NS}	08.90 ± 0.50 ^b	14.87 ± 6.11 ^a
IC + HMMN	200	43.73 ± 1.60 ^{NS}	18.58 ± 6.01 ^c	21.71 ± 7.24 ^a
IC + Diclofenac	10	44.68 ± 0.99 ^a	30.80 ± 0.01 ^c	32.75 ± 3.19 ^b

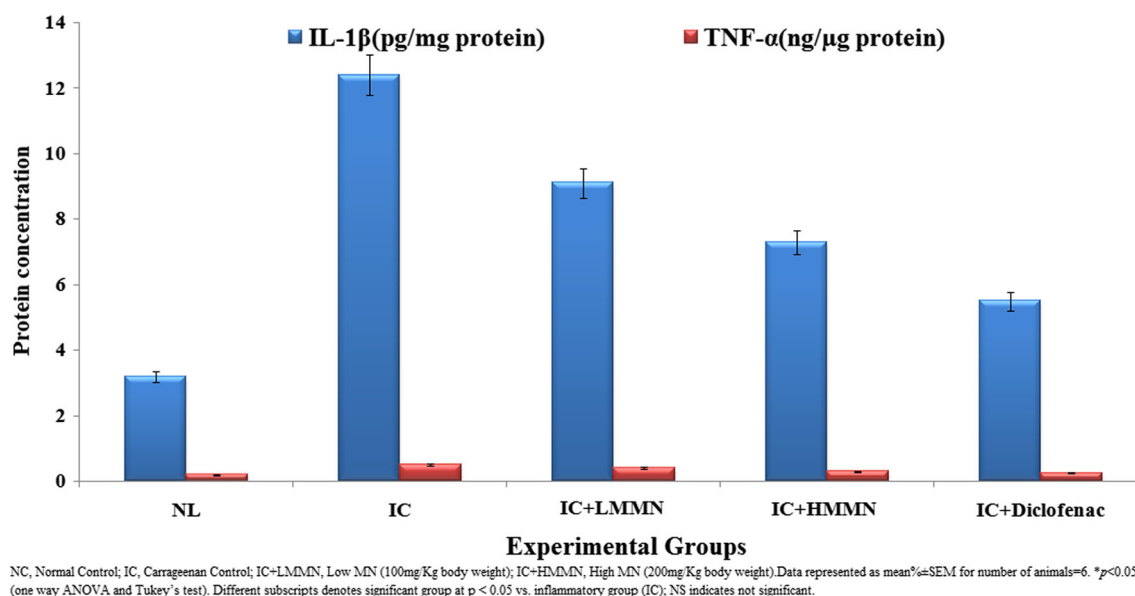
NC, Normal Control; IC, Carrageenan Control; IC + LMMN, Low MN (100 mg/Kg body weight); IC + HMMN, High MN (200 mg/Kg body weight). Data represented as mean% ± SEM for number of animals = 6. * $p < 0.05$ (one way ANOVA and Tukey's test). Different subscripts denotes significant group at $p < 0.05$ vs. inflammatory group (IC); NS indicates not significant

vealed that MMN significantly ($p < 0.01$) produced a dose-dependent inhibition of paw edema which was comparable to the effect of a known anti-inflammatory drug, diclofenac in the experimental rats at fourth hour. The reduction in carrageenan-induced paw edema by HMMN after 4 h was 21.71 % while edema reduction by the standard drug, diclofenac (10 mg/kg) was 32.75 % (Table 2). Our in vivo results were in strongly proof and in accordance to previous studies in other plant species (Usha et al. 2014; Mohanthy et al. 2015).

3.4.2 Effect of MMN on various cytokine levels

The roles of natural product obtained from plants have been well documented in treatment of both acute and chronic inflammatory diseases. Cytokines, small glyco-

proteins play crucial role in elicitation of inflammatory reaction. Each of the cytokines (IL-1 β and TNF- α) tested here are critically involved in controlling innate as well as adaptive inflammatory response (Loram et al. 2007). The level of these cytokines should be suppressed to reduce the severity of the inflammatory reaction. In the present study, the regulation of IL-1 β and TNF- α in serum were determined to access the anti-inflammatory activity of both HMMN and LMMN in comparison to the reference drug (diclofenac) and control. It is evident from the Fig. 2 that inflammatory reaction induced by carrageenan significantly increased concentration of IL-1 β and TNF- α ($p < 0.05$) in serum. Though both the doses were found to be effective in reducing the levels of cytokines in the serum significantly as compared to the inflammation-induced groups, HMMN was more effective.

**Fig. 2** Effects of methanolic extracts of *M. nagi* leaves on the levels of cytokines in experimental rats

4 Conclusions

In conclusion, we can infer that the methanolic extract of *Myrica nagi* leaves is a potential source of natural antioxidants. In addition to this our findings provide scientific rationale for the folk use of *Myrica nagi* as an analgesic and anti-inflammatory shrub. However, further study is needed in order to understand the precise mechanism and be totally authentic. Phytochemical characterization and pharmacodynamics studies are also proposed to isolate the active constituents and establish the mechanism of action.

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Compliance with ethical standards

Conflict of interest Authors have no conflict of interest.

Ethical approval All applicable international, national, and/or institutional guidelines for the care and use of animals were followed. All procedures performed in studies involving animals were in accordance with the ethical standards of the institution or practice at which the studies were conducted.

References

- Ahmed F, Hossain MH, Rahman AA, Shahid IZ (2006) Antinociceptive and sedative effects of the bark of *Cerbera odollam* Gaertn. Orient Pharm Exp Med 6:344–348
- Comalada M, Camuesco D, Sierra S, Ballester I, Xaus J, Gálvez J, Zarzuelo A (2005) *In vivo* quercitrin anti-inflammatory effect involves release of quercetin, which inhibits inflammation through down-regulation of the NF- κ B pathway. Eur J Immunol 35:584–592
- Duarte IDG, Nakamura M, Ferreira SH (1988) Participation of the sympathetic system in acetic acid-induced writhing in mice. Braz J Med Biol Res 21:341–343
- Goyal AK, Middha SK, Sen A (2010) Evaluation of the DPPH radical scavenging activity, total phenols and antioxidant activities in Indian wild *Bambusa vulgaris* “Vittata” methanolic leaf extract. J Nat Pharm 1(1):34–39
- Goyal AK, Basistha BC, Sen A, Middha SK (2011a) Antioxidant profiling of *Hippophae salicifolia* growing in sacred forests of Sikkim, India. Funct Plant Biol 38:697–701
- Goyal AK, Middha SK, Sen A (2011b) *In vitro* antioxidative profiling of different fractions of *Dendrocalamus strictus* (Roxb.) Nees leaf extracts. Free Rad Antioxid 1:42–48
- Goyal AK, Mishra T, Bhattacharya M, Kar P, Sen A (2013) Evaluation of phytochemical constituents and antioxidant activity of selected actinorhizal fruits growing in the forests of Northeast India. J Biosci 38(4):797–803
- Khan MY, Sagrawat H, Upmanyu N, Siddique S (2008) Anxiolytic properties of *Myrica nagi* bark extract. Pharm Biol 46:757–761
- Loram LC, Fuller A, Fick LG, Cartmell T, Poole S, Mitchell D (2007) Cytokine profiles during carrageenan-induced inflammatory hyperalgesia in rat muscle and hind paw. J Pain 8:127–136
- Mannello F (2006) Natural Bio-Drugs as matrix metalloproteinase inhibitors: new perspectives on the horizon. Recent Pat Anti-Cancer Drug Discov 1:91–103
- Middha SK, Bhattacharjee B, Saini D, Baliga MS, Nagaveni MB, Usha T (2011) Protective role of *Trigonella foenum graecum* extract against oxidative stress in hyperglycemic rats. Eur Rev Med Pharmacol Sci 15(4):427–435
- Mohanty SK, Swamy MK, Middha SK, Prakash L, Subbanarashiman B, Maniyam A (2015) Analgesic, anti-inflammatory, anti-lipoxygenase activity and characterization of three bioactive compounds in the most active fraction of *leptadenia reticulata* (Retz.) Wight & Arn.—a valuable medicinal plant (2015 summer). Iran J Pharm Res 14(3): 933–942
- Panthari P, Kharkwal H, Kharkwal H, Joshi DD (2012) *Myrica nagi*: a review on active constituents, biological and therapeutic effects. Int J Pharm Pharm Sci 4:38–42
- Patel T, Rajshekar C, Parmar R (2011) Mast cell stabilizing activity of bark of *Myrica nagi*. J Pharm Stud Res 2:1–6
- Silva BJC, Seca AML, Barreto MC, Pinto DCGA (2015) Recent breakthroughs in the antioxidant and anti-inflammatory effects of *Morella* and *Myrica* species. Int J Mol Sci 16(8):17160–17180
- Usha T, Middha SK, Bhattacharya M, Lokesh P, Goyal AK (2014) Rosmarinic acid, a new polyphenol from *Baccaurea ramiflora* Lour. leaf: a probable compound for its anti-inflammatory activity. Antioxidants 3(4):830–842
- Vedhanayaki G, Shastri GV, Kuruvilla A (2003) Analgesic activity of *Piper Longum* Linn. root. Indian J Exp Biol 41:649–651
- Viana AF, Maciel IS, Motta EM (2011) Antinociceptive activity of *Trichilia catigua* hydroalcoholic extract: new evidence on its dopaminergic effects. Evid Based Complement Alternat Med 2011:1–8
- Winter CA, Risley EA, Nuss GW (1962) Carrageenan induced oedema in hind paw of rat as an assay for anti-inflammatory drugs. P Soc Exp Biol Med 111:544–547
- Yanthan M, Misra AK (2013) Molecular approach to the classification of medicinally important actinorhizal genus *Myrica*. Indian J Biotechnol 12:133–136
- Zakaria ZA, Abdul Gani ZDF (2008) Antinociceptive, anti-inflammatory, and antipyretic properties of an aqueous extract of *Dicranopteris linearis* leaves in experimental animal models. J Nat Med 62:179–187
- Zulfiker AHM, Rahman MM, Hossain MK, Hamid K, Mazumder MEH, Rana MS (2010) *In vivo* analgesic activity of ethanolic extracts of two medicinal plants-*Scoparia dulcis* L. and *Ficus racemosa* Linn. Biol Med 2(2):42–48